

Introducing Micro Clot Check on ABL90 FLEX and FLEX PLUS*

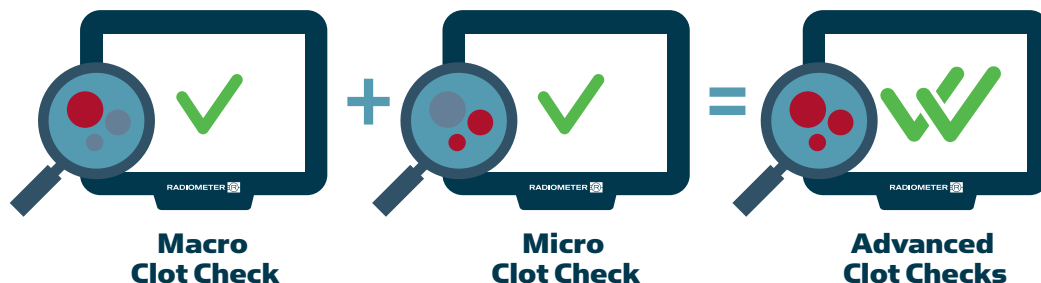
Even with a high attention to the crucial and sensitive pre-analytical phase, clotted samples are found to be one of the most common pre-analytical issues. Building on Radiometer's Advanced Clot Checks and Removal technology [1,2,3]. Micro Clot Check is a new and additional clot management system that checks for clots that are too small to be detected by analyzer's flow sensors.

Micro Clot Check now expands clot management beyond flow monitoring to further reduce the risk that a clot affects the analyzer's performance. While flow monitoring with Macro Clot Check effectively detects larger clots that disrupt flow, Micro Clot Check performs additional and continuous sensor response analysis to seek and remove smaller clots.

If undetected, micro clots could become lodged on the analyzer's sensors and potentially affect the next measurement, calibration or quality control (QC).

If a clot is detected, the analyzer automatically initiates a back flush procedure to reverse the flow and clear the clot. Once this correction has been completed, its effectiveness is verified with a separate check and these actions are documented in the activity log to support compliance.

As part of Radiometer's Automatic Quality Management (AQM) system on the ABL90 FLEX and FLEX PLUS analyzer, Micro Clot Check reduces the risk of undetected clots and helps give caregivers confidence in the measurements they depend on when taking care of their patients.



Micro Clot Check: Setting new standards with AQM

Micro Clot Check is the latest addition to Radiometer's AQM system on the ABL90 FLEX and FLEX PLUS. AQM automatically performs a combination of prescheduled and continuous real-time checks of the analyzer, initiates corrective actions without requiring manual intervention, and documents activities to support compliance.

AQURE POC IT solution supports compliance

AQURE point-of-care IT solution gives you access to QC results and reports for inspection readiness. In addition, you can manage your analyzers, operators, and their competencies, directly from your desk.



Detection

- QC checks
- Calibration checks
- Analysis checks
- System checks
- Clot checks

Correction

- Automatic corrective actions
Reduce downtime
- Intelligent troubleshooting
Increase consumable utilization and uptime

Reporting

- Lab standard
Levey-Jennings Plots, QC statistics, Activity log

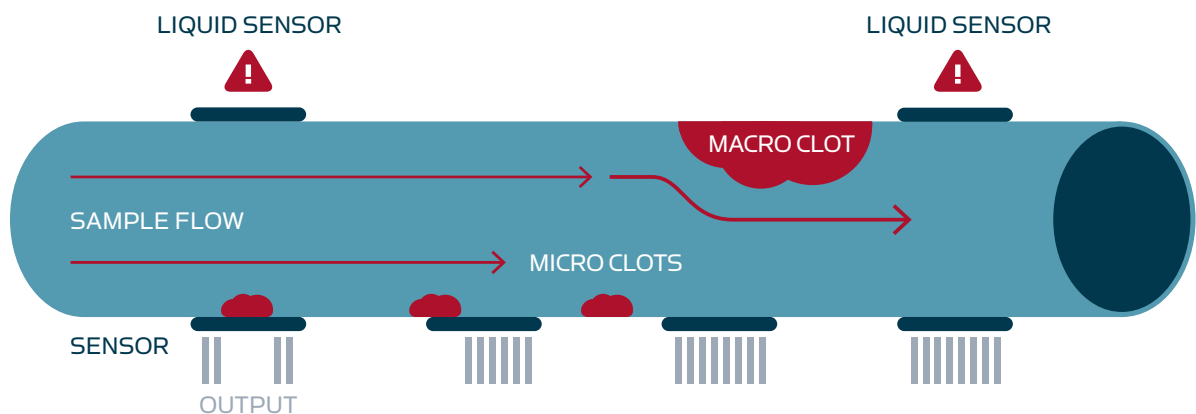
* Micro Clot Check is not available when ABL 90 FLEX PLUS is configured with creatinine and urea.

The challenge of clots in POC blood gas testing

In critical care, accurate blood gas analyzer measurements are important for making informed treatment decisions fast. But even when the most rigorous preanalytical precautions are followed,

blood clots can still be introduced into an analyzer via a sample, and may potentially affect measurement, calibration or QC. Macro Clot Check and Micro Clot Check work in tandem to reduce this risk.

MACRO CLOT CHECK uses advanced flow monitoring to detect macro clots that disrupt flow and can affect sensor performance.



MICRO CLOT CHECK uses continuous sensor response analysis to detect and remove micro clots that do not affect flow but which can attach on or near sensors and potentially affect sensor performance.

Advanced Clot Checks and Removal

Macro Clot Check and Micro Clot Check combine to deliver Advanced Clot Checks and Removal on the ABL90 FLEX and FLEX PLUS. Micro Clot Check integrates three novel clot detection methods.



1 BACKLOG CLOT DETECTION

Detects if a micro clot has attached to a sensor after a measurement.

2 IDLE CLOT DETECTION

Detects clots that affect sensors over time, typically small clots on or near sensors.

3 AIR/LIQUID CLOT DETECTION

Detects clots by comparing actual and expected sensor response when measuring on air or liquid.

1. Giavarina D., Lippi G. Blood venous sample collection: Recommendations overview and a checklist to improve quality. Clin Biochem 2017; 50:568–573.
2. Guimarães A. C., Wolfart M., Brisolara M. L. L., Dani C. Causes of rejection of blood samples handled in the clinical laboratory of a University Hospital in Porto Alegre. Clin Biochem 2012; 45:123–126.
3. Bhat V., Tiwari M., Chavan P., Kelkar R. Analysis of laboratory sample rejections in the pre-analytical stage at an oncology center. Clinica Chimica Acta 2012; 413:1203–1206.